DMAIC · REAL DATA · BPI CHALLENGE 2020

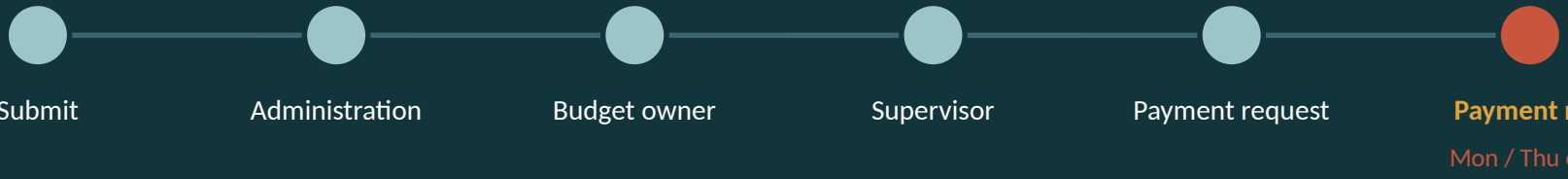
Approval Workflow Improvement

Why 1 in 10 expense claims waits 20+ days to be paid, and three fixes that meet every target

20.25

days

for the slowest 1 in 10 claims to be paid
(P90, 7,903 paid claims, 2018)



Vimlesh Sonawane · November 2026

Agenda

Click a phase to jump to it. Every slide has a ← Agenda link.



Define

Charter, scope, roles

Slide 4 →



Measure

Baseline KPIs and patterns

Slide 5 →



Analyze

Process map, waits, rework, root causes

Slide 7 →



Improve

Fixes, to-be process, what-if scenarios

Slide 11 →



Control

Roadmap and control plan

Slide 15 →

Summary →

Before vs after →

Lessons →

The ask →

Three fixes together meet all three targets

← Agenda



Problem

1 in 10 paid claims takes over 20.25 days. 13.14% are rejected at least once.



Causes

Payments run only Mon/Thu. Approval waits have long tails. Rejections cause rework.



Fix

Daily payment runs, a checklist with an amount > 0 rule, reminders after 3 days.

Modelled effect of all three fixes (S6, what-if on 2018 data)

P90 days	Rejected	Payment wait
20.25 → **8.74**	13.14% → **7.30%**	3.21 d → **1.00 d**

Real data, run as a DMAIC project

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Define	Measure	Analyze	Improve	Control
Charter · stakeholders · RACI	Clean data · baseline KPIs · Tableau	As-is map · bottlenecks · fishbone	Impact/effort · to-be · scenarios	Roadmap · control plan · deck
D1–D2	D3	D4–D5	D6	D7–D8

The data

BPI Challenge 2020 Domestic Declarations: 10,500 claims, 56,437 events. Verified against the publisher's checksum. 2018 = main view.

The tools

Python · Tableau · Smartsheet (schedule, RAID, roadmap) · Asana (tasks, change requests, status) · Miro (maps)

Baseline: the typical claim is fine, the tail is not

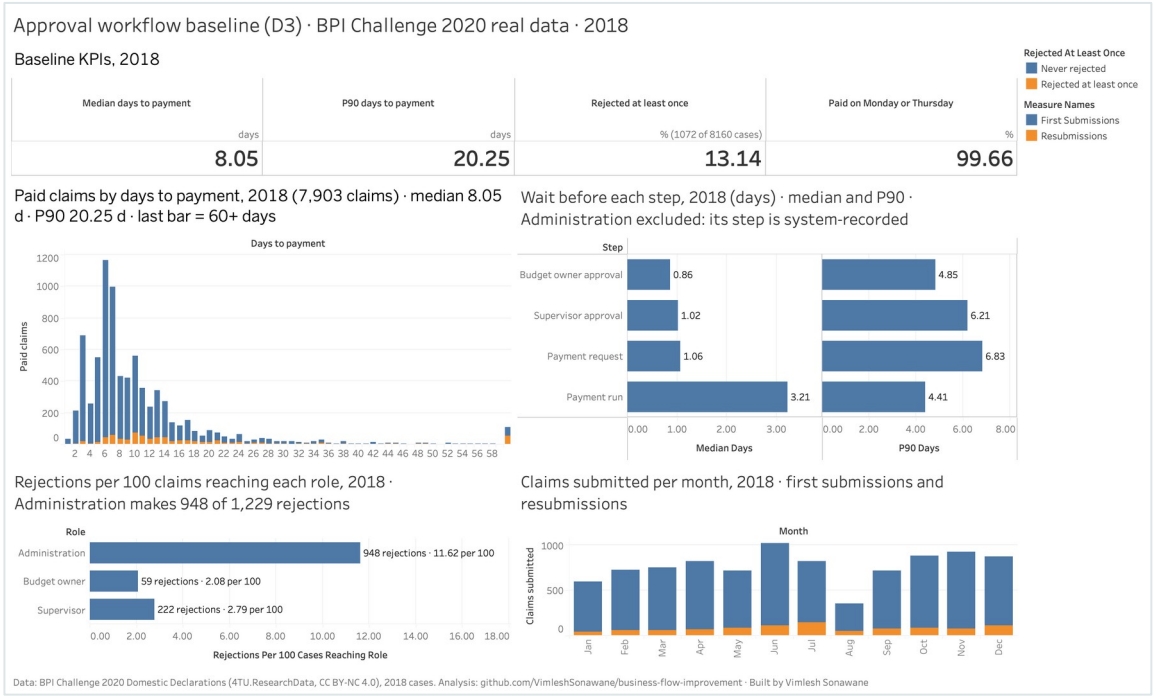
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8.05 d — Median days to pay

20.25 d — P90 days to pay

13.14% — Rejected at least once

99.66% — Paid on Mon or Thu

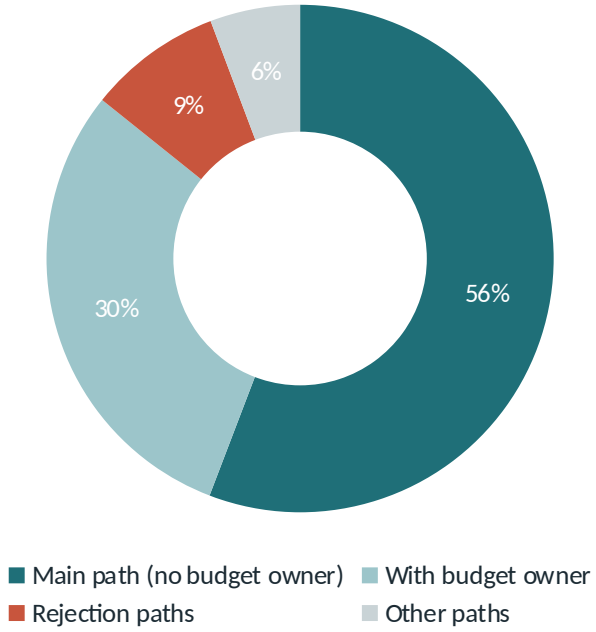


Live Tableau dashboard: click the image to open it

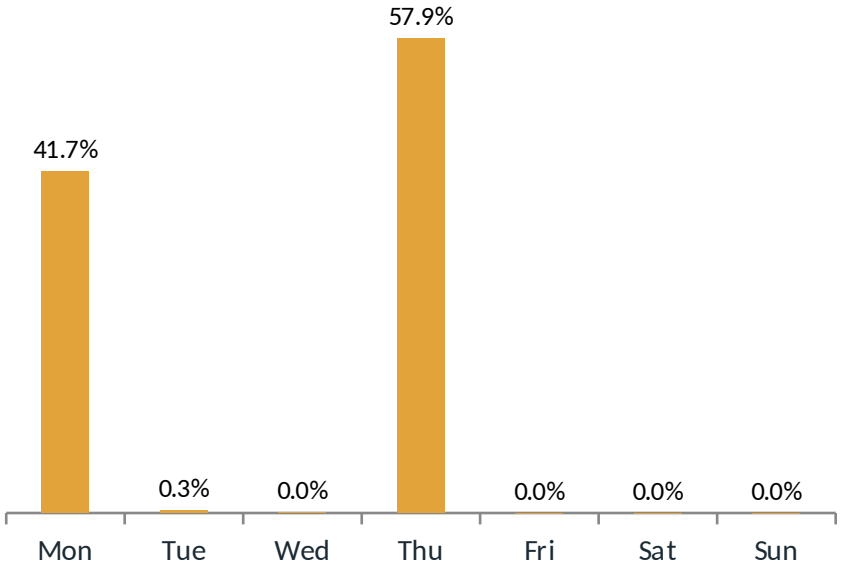
One main path; payments only twice a week

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How 2018 claims flow (8,260 claims)

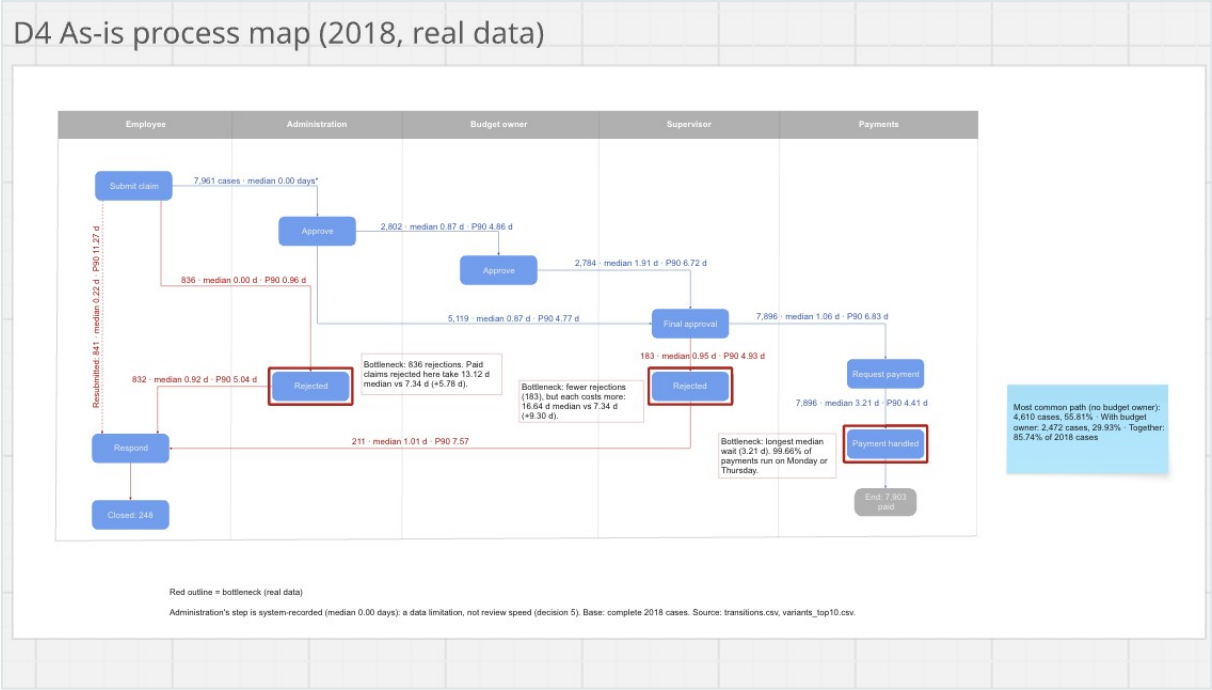


Payments by weekday (7,903 payments)



As-is map: three bottlenecks in red

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B1 Payment run

Longest median wait, 3.21 d; Mon/Thu batches

B2 Administration rejects

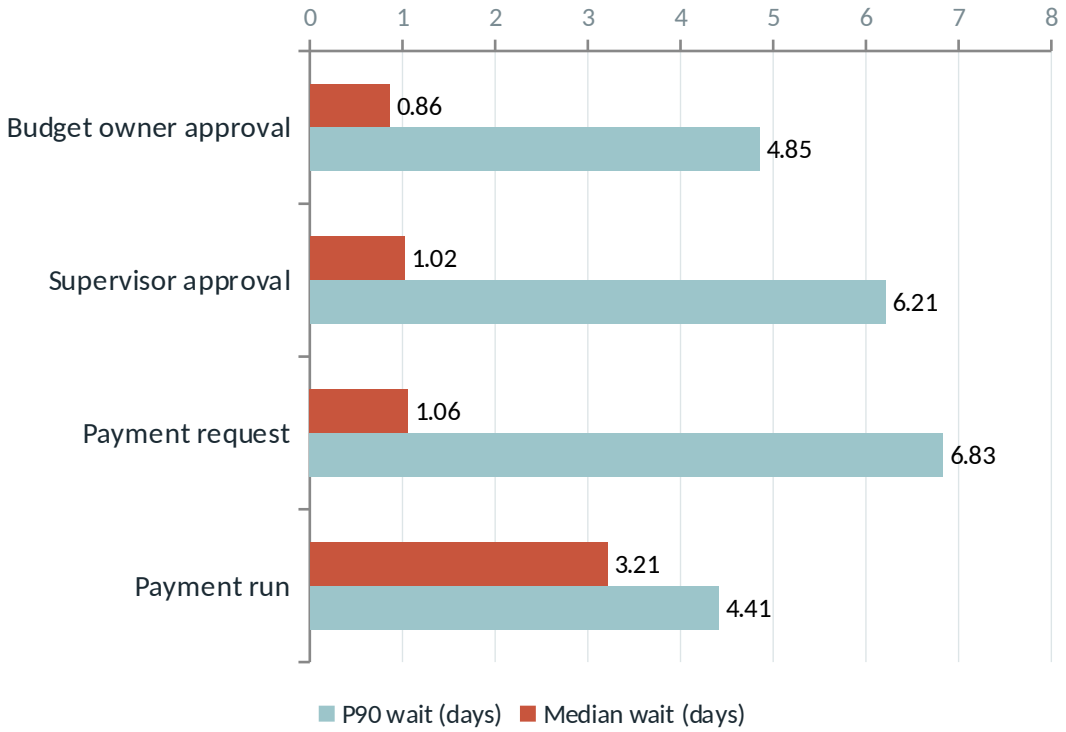
836 rejections; +5.78 d when it happens

B3 Supervisor rejects

Only 183, but +9.30 d each

Batching slows every claim; approvals create the tail

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Median: payment run

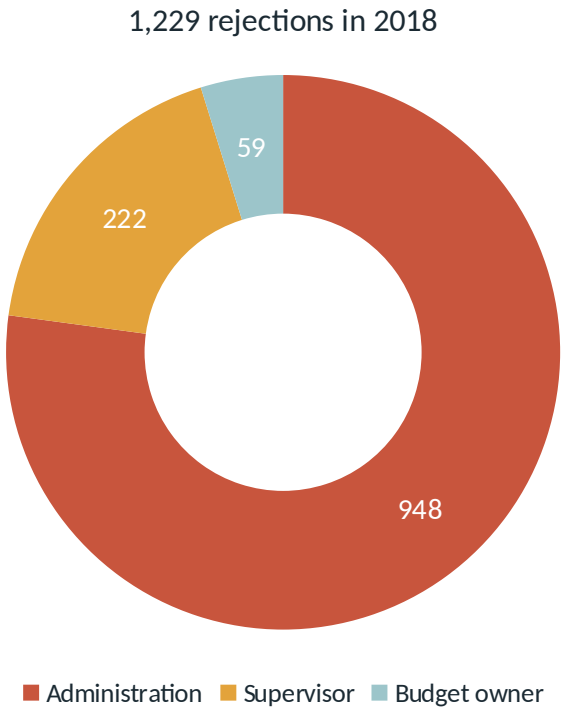
3.21 d, because payments run only twice a week.

P90: approvals

Up to 6.83 d: most claims move fast, a few sit for a week.

Rework: Administration makes 3 of 4 rejections

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1.8×

Longer to pay when rejected (13.21 vs 7.34 d)

+9.30 d

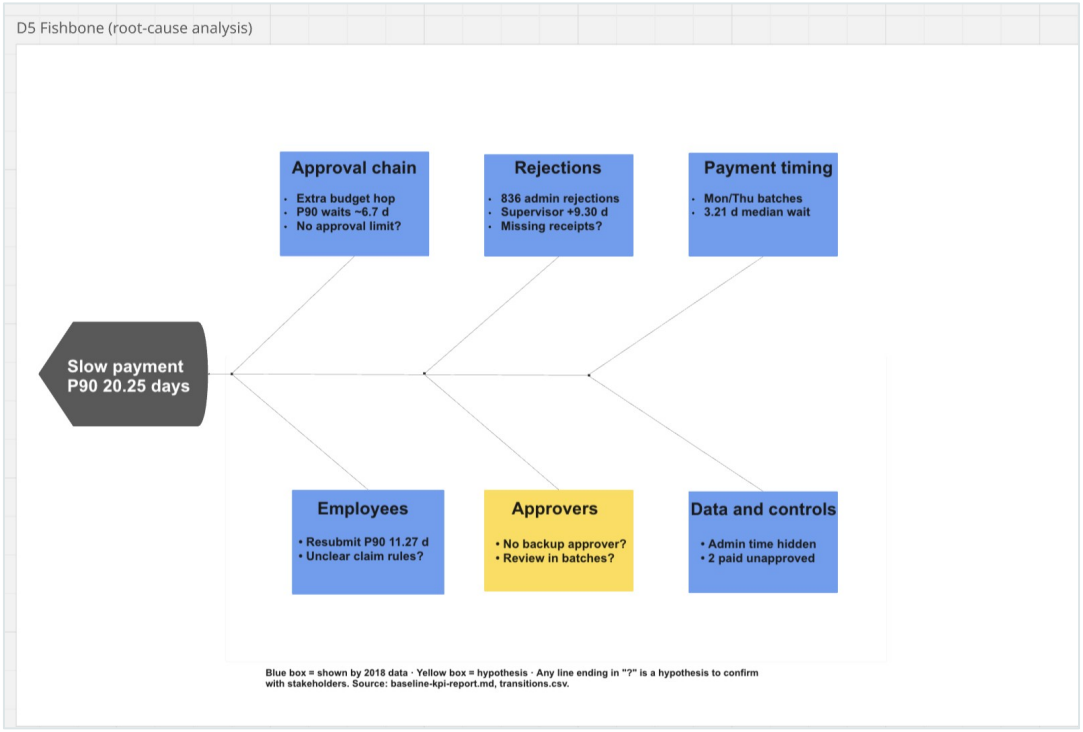
Extra median days when a supervisor rejects

253 of 1,072 rejected claims had an amount of zero

Every one was rejected and none was paid. An amount > 0 check at submission would stop all of them.

Root causes: proven by data vs still to test

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How to read it

- Blue boxes are shown by 2018 data.
- The yellow box is a hypothesis to test with stakeholders.
- Any line ending in "?" is still a hypothesis.
- Why claims are rejected isn't in the data, so that needs interviews.

Eight fixes, scored for impact and effort

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QUICK WINS

- **F2 Claim checklist**
- **F3 Approval reminders**
- F4 Backup approver

MAJOR PROJECTS

- **F1 Daily payment runs**
- F8 Payment control

FILL-INS

- F6 Reason codes
- F7 Log admin time

NOT NOW

- F5 Skip budget step

Impact ↑ Effort →

Top 3 for the business case

F1 Daily payment runs
→ payment wait ≤ 1 d

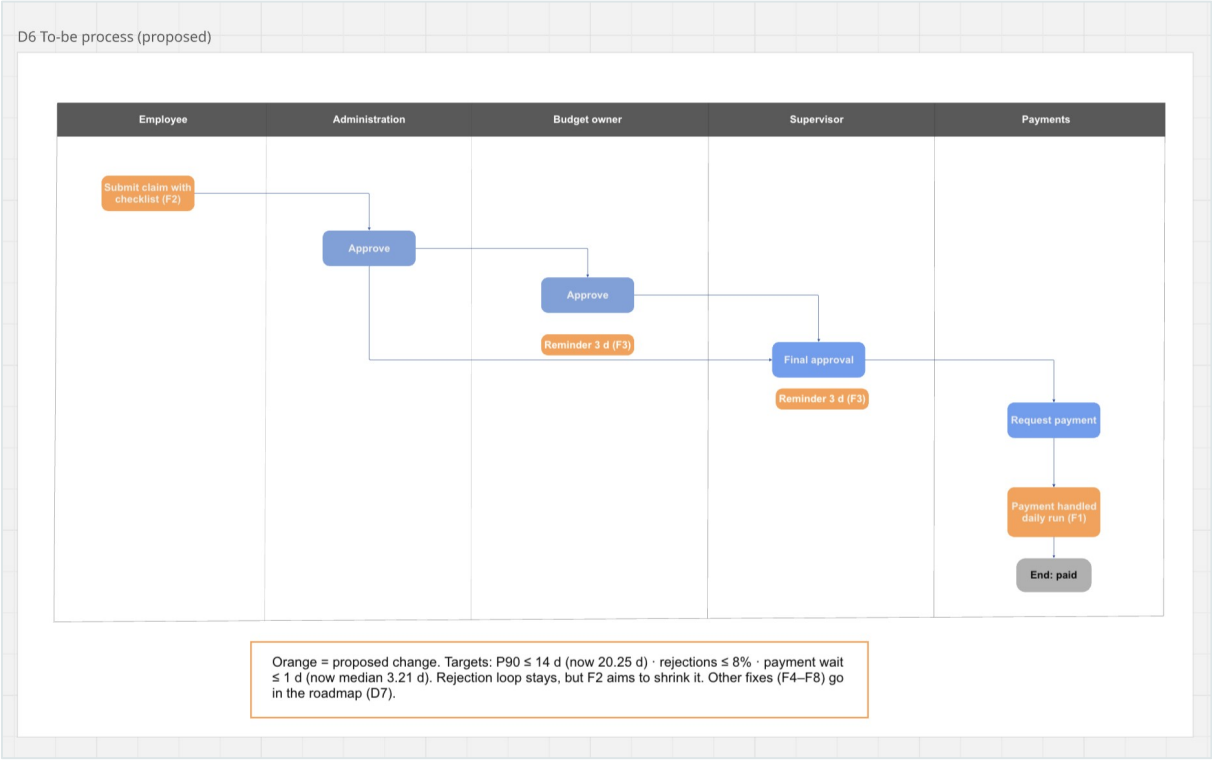
F2 Checklist + amount > 0
→ rejections ≤ 8%

F3 Reminders after 3 days
→ P90 ≤ 14 d

Placement is a PM judgment, to confirm with the process owner.

To-be: the same flow, with three proposed changes

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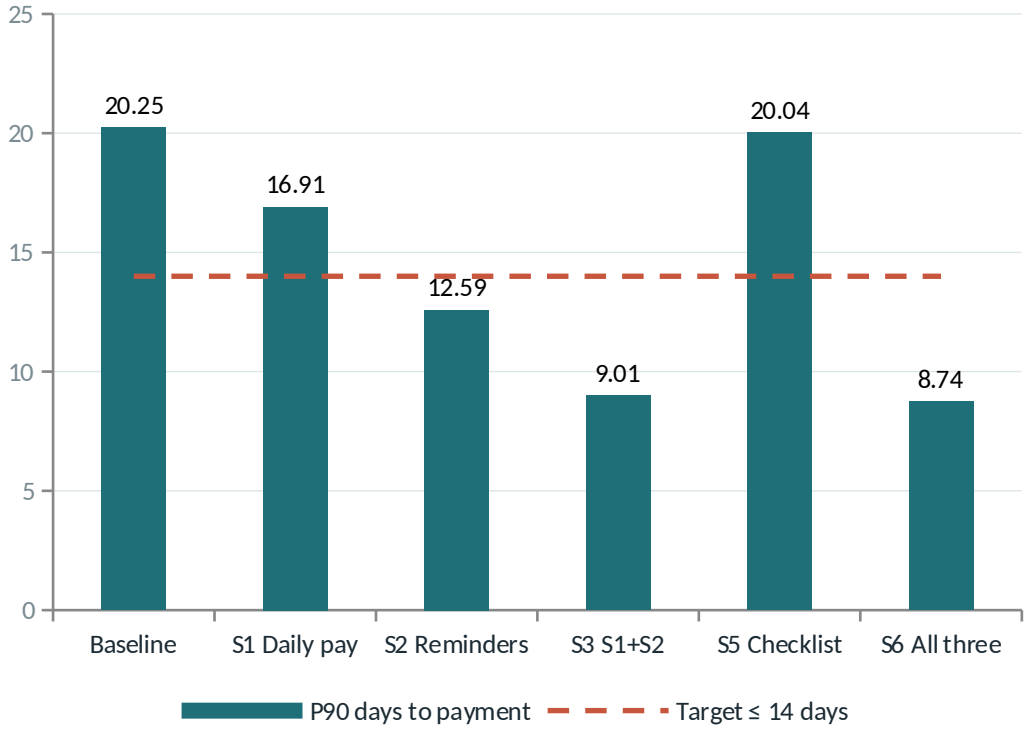
F2
Checklist at submission

F3
Reminders after 3 days

F1
Daily payment run

Scenarios: all three fixes meet every target

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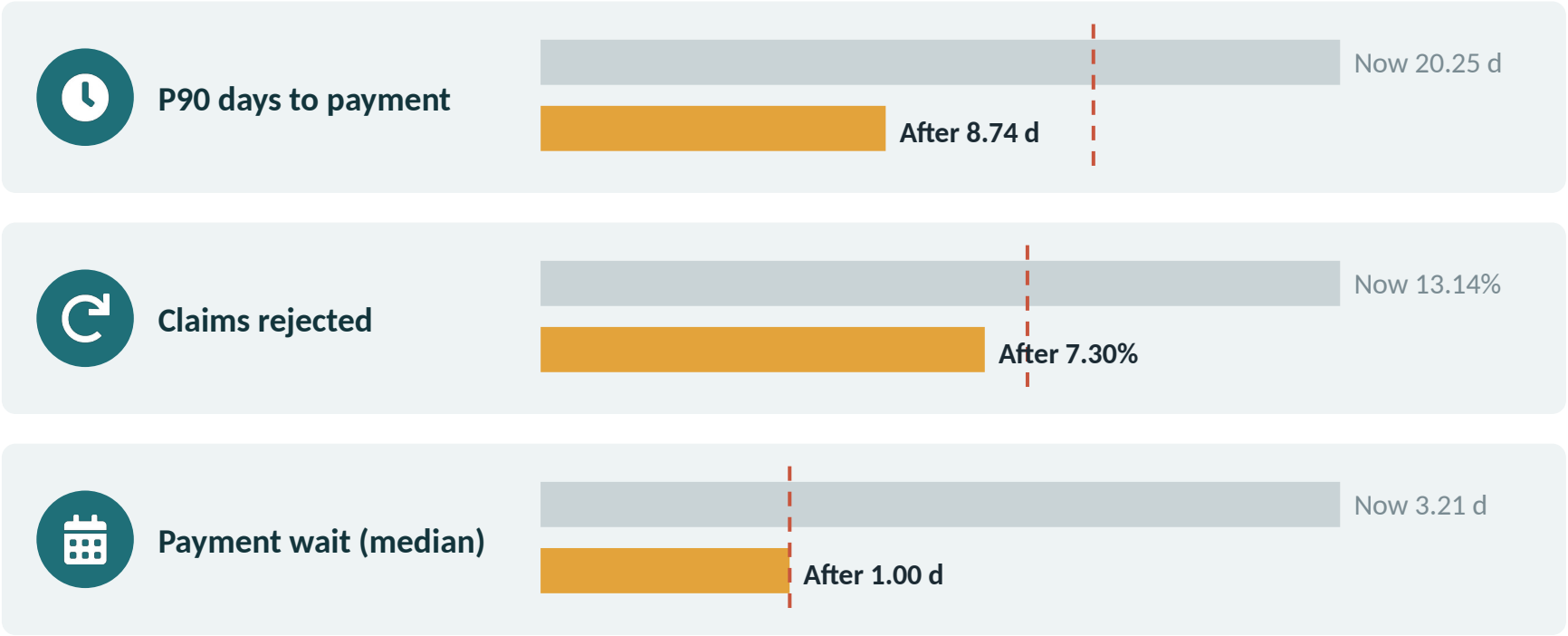


Scenario	P90 ≤ 14 d	Rej. ≤ 8%	Pay ≤ 1 d
S1 Daily pay	–	–	✓
S2 Reminders	✓	–	–
S3 S1+S2	✓	–	✓
S5 Checklist	–	✓	–
S6 All three	✓	✓	✓

Estimates on 2018 data. The 40% cut in Administration rejections is assumed; the zero-amount effect is observed.

Before vs after: all three charter targets met (S6)

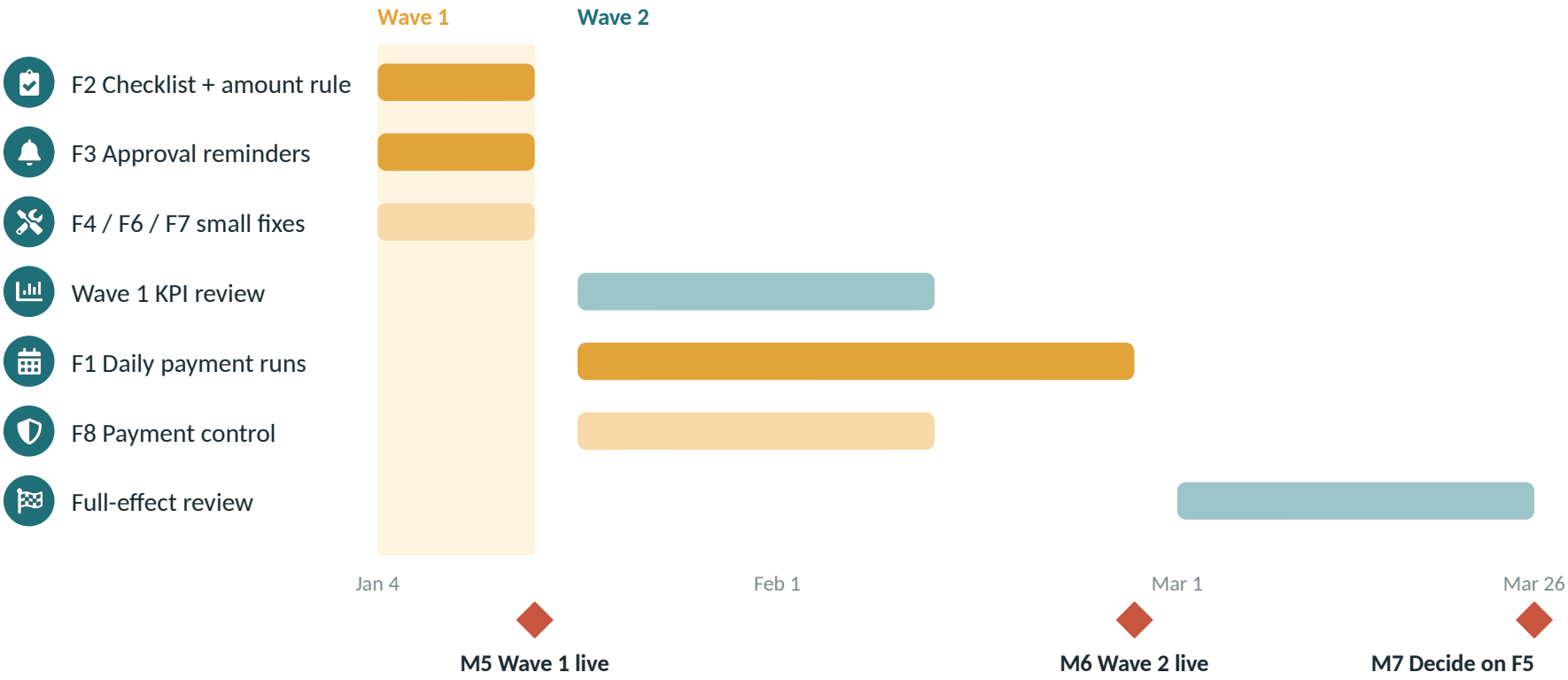
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Dashed red line = charter target. What-if estimates on 2018 data, not results.

Roadmap: quick wins in Jan, payments by Feb

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Control plan: how the gains are kept

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KPI	Now	Target	Alert if	Owner
P90 cycle time	20.25 d	≤ 14 d	> 14 d two weeks running	Process owner
Rejection rate	13.14%	≤ 8%	> 9% in a month	Process owner
Zero-amount submissions	253	0	Any	IT / workflow admin
Payment wait (median)	3.21 d	≤ 1 d	> 1.5 d in a week	Payments team
Paid without final approval	2	0	Any	Finance



Review rhythm

Weekly for 4 weeks after each wave, then monthly. Sponsor review quarterly.

Full plan: 10 KPIs in Smartsheet.

What these numbers are, and what I learned

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Limits

- Scenarios are what-if estimates on 2018 data, not results.
- Fixes were not implemented; stakeholders were role-played.
- Administration's review time is system-recorded, so it can't be measured.
- 2 claims paid without final approval are flagged, not judged.

Lessons

- The data changed the plan: I expected daily payments to fix the P90, but reminders do.
- Checking a coincidence (two counts of 253) uncovered the zero-amount finding.
- Labelling facts vs hypotheses kept every claim defensible.

The ask: approve Wave 1 for January



Checklist + amount > 0 rule



Approval reminders



Backup approvers



Reason codes + admin-time log

Low effort, starts the rejection and P90 improvements. Wave 2 (daily payment runs, payment control) follows the Wave 1 review.

GitHub: code and analysis →

Tableau: live baseline dashboard →

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